

Chapter 1

Section 1 - Basic Usage

1.1 Section 1.1 - Intro to Logic

1.1.1 Propositions

Definition 1 (Proposition)

A **proposition** is a statement that is either

Example 1 (Examples with Propositions)

Determine if the following are propositions

1. It's all coming back to me now

Solution:

yes

2. Hello.

Solution:

No

3. $x + 3 = 10$

Solution:

No, since we don't know x

4. In a right triangle, $a^2 + b^2 = c^2$

Solution:

No, since a, b, c are not assigned anything

5. In a right triangle having legs of length a and b , and hypotenuse of c , then $a^2 + b^2 = c^2$

Solution:

Yes, since a, b, c are properly specified

6. There are 75 people in this room or the sky is purple

Solution:

Yes

Note

7 is a **compound proposition**

Definition 2 (Compound Proposition)

A **compound proposition** is a proposition that contains two or more **atomic** propositions. In the previous example, the two are:

1. There are 75 people in this room
2. The sky is purple

Definition 3 (Atomic Proposition)

An **atomic proposition** is a proposition that cannot be broken down further into separate propositions

Note

We denote separate propositions as: p, q, r

1.1.2 Logical Operations (connectives)

Name	Notation	Phrasing	Truth Value
Disjunction (or)	$p \vee q$	"p or q"	True when p, q or both are TRUE
Conjunction (and)	$p \wedge q$	"p and q"	True when both are TRUE
Negation (not)	$\neg p$	"not p"	The opposite truth value of p
Exclusive Or (XOR)	$p \oplus q$	"p or q, but not both"	TRUE when only one is TRUE

Here are the truth tables of each operation

p	q	$p \vee q$	$p \wedge q$	$\neg p$	$p \oplus q$
T	T	T	T	F	F
T	F	T	F	F	T
F	T	T	F	T	T
F	F	F	F	T	F

Example 2 (Logical Operators Examples)

Let p be the proposition "a is odd" and q be "a is even" where $a = 3$

p : 3 is odd = **TRUE**

q : 9 is even = **FALSE**

Write the following Propositions:

1. $\neg p$ (3 is even)

Solution:

FALSE

2. $\neg q$ (9 is odd)

Solution:

TRUE

3. $p \vee q$ (3 is odd or 9 is even)

Solution:

TRUE since 3 is odd

4. $p \wedge q$ (3 is odd and 9 is even)

Solution:

FALSE since 9 is not even

5. $p \wedge \neg q$ (3 is odd and 9 is odd)

Solution:

TRUE since both 3 is odd and 9 is odd

1.1.3 Conditionals

Name	Notation	Phrasing	Truth Value
Conditional	$p \Rightarrow q$	"p implies q" or "if p then q"	Always True except when $T \Rightarrow F$
Biconditional	$p \Leftrightarrow q$	"p if and only if q"	True when p and q are the same value

Here are the truth tables

p	q	$p \Rightarrow q$	$p \Leftrightarrow q$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	T

1.1.4 Logical Transforms

Name	Original	Output
Converse (swap)	$p \Rightarrow q$	$q \Rightarrow p$
Inverse (negate)	$p \Rightarrow q$	$\neg p \Rightarrow \neg q$
Contrapositive (converse + inverse)	$p \Rightarrow q$	$\neg q \Rightarrow \neg p$

Here are the truth tables

p	q	$p \Rightarrow q$	$\neg q$	$\neg p$	$\neg q \Rightarrow \neg p$
TRUE	TRUE	TRUE	FALSE	FALSE	TRUE
TRUE	FALSE	FALSE	TRUE	FALSE	FALSE
FALSE	TRUE	TRUE	FALSE	TRUE	TRUE
FALSE	FALSE	TRUE	TRUE	TRUE	TRUE

Proposition 1 (Contrapositive Correlation)

From the truth table, we see that the converse is logically equivalent to the original expression, so

$$p \Rightarrow q \equiv \neg q \Rightarrow \neg p$$

Definition 4 (Logical Equivalence)

Two propositions are **logically equivalent** if they have the same truth tables

Example 3 (Logical Transforms Examples)

Let p be the proposition "it's cold outside" and q be "it's snowing outside"

1. $p \implies q$ (if it's cold outside, then it's snowing outside)

Solution:

FALSE

2. $q \implies p$ (converse) (if it's snowing outside, then it's cold outside)

Solution:

TRUE

3. $\neg p \implies \neg q$ (inverse) (if it's not cold outside, then it's not snowing)

Solution:

TRUE

4. $\neg q \implies \neg p$ (contrapositive) (if it's not snowing outside, then it's not cold outside)

Solution:

FALSE

Example 4 (More Examples with Logical Transforms)

Let p be the proposition "The triangle is equilateral" and q be "The triangle is equiangular"

1. $p \implies q$ (if the triangle is equilateral, then triangle is equiangular)

Solution:

TRUE

2. $q \implies p$ (converse) (if the triangle is equiangular, then the triangle is

equilateral)

Solution:

TRUE

3. $\neg p \implies \neg q$ (inverse) (if the triangle is not equiangular, then the triangle is not equilateral)

Solution:

TRUE

4. $\neg q \implies \neg p$ (contrapositive) (if the triangle is not equiangular then the triangle is not equilateral)

Solution:

TRUE

Example 5 (Truth Table Examples)

1. $\neg p \vee q$

p	q	$\neg p$	$\neg p \vee q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

2. $(p \implies r) \wedge (q \implies r)$

im too lazy to fill this out

1.3 Section 1.3 - Logical Equivalence

Recall 1 (Logical Equivalence)

Recall that two propositions are **logically equivalent** if they have the same truth table

1.3.1 Testing for Equivalence using Truth Tables

Example 6

Determine if the following are logically equivalent

1. $p \implies q$ and $\neg p \vee q$

Solution:

p	q	$p \implies q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

$$\therefore p \implies q \equiv \neg p \vee q$$

2. $\neg(p \wedge q)$ and $\neg p \vee \neg q$

Solution:

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

$$\therefore \neg(p \wedge q) \equiv \neg p \vee \neg q$$

3. $(p \implies q) \vee (q \implies r)$ and $p \implies (q \vee r)$

Solution:

p	q	r	$p \implies r$	$q \implies r$	$(p \implies r) \vee (q \implies r)$	$q \vee r$	$p \implies (q \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	T
T	F	T	T	T	T	T	T
T	F	F	F	T	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	F	T	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	F	T

$$\therefore (p \implies q) \vee (q \implies r) \not\equiv p \implies (q \vee r)$$

So far we know the following:

- $p \implies q \equiv \neg p \vee q$
- DeMorgan's Law
 - $\neg(p \vee q) \equiv \neg p \wedge \neg q$
 - $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- Commutative (swap places)
 - $p \vee q \equiv q \vee p$
 - $p \wedge q \equiv q \wedge p$
- Associative (swap grouping)
 - $(p \vee q) \vee r \equiv p \vee (q \vee r)$
 - $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- Distributive (distributing operations)
 - $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
 - $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- Identity
 - $p \wedge T \equiv p$
 - $p \vee F \equiv p$
- Domination Laws
 - $p \vee T \equiv T$
 - $p \wedge F \equiv F$
- Impotent Laws
 - $p \vee p \equiv p$

$$- p \wedge p \equiv p$$

Definition 5 (Tautology and Contradiction)

A proposition that is always **TRUE** is a **tautology**

A proposition that is always **FALSE** is a **contradiction**

Note

When proving tautologies or contradictions, you could either:

1. Use Truth Tables
2. Use the Laws from earlier to reduce a compound proposition into **TRUE** or **FALSE**

But it's advised to use Truth Tables, since it's easy to get lost in your work when using the Laws

Also, ALWAYS start by **converting conditionals**

Example 7 (Examples of Tautologies and Contradictions)

Determine if the following are tautologies, contradictions, or neither

1. $(p \wedge q) \implies q$

Proving with Truth Tables

p	q	$p \wedge q$	q	$(p \wedge q) \implies q$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	F	T

$\therefore (p \wedge q) \implies q \equiv \text{tautology}$

Proving with Logical Equations

$$\begin{aligned}
 (p \wedge q) \implies q &\equiv \neg(p \wedge q) \vee q && \text{(Convert Conditional)} \\
 &\equiv (\neg p \vee \neg q) \vee q && \text{(DeMorgan's Law)} \\
 &\equiv \neg p \vee (\neg q \vee q) && \text{(Associative)} \\
 &\equiv \neg p \vee T && \\
 &\equiv T && \text{(Domination)}
 \end{aligned}$$

$\therefore (p \wedge q) \implies q \equiv \text{tautology}$

2. There's another example but I'm too lazy

Note

When proving the **logical equivalence** of two propositions, start with the **MORE COMPLICATED** proposition

Example 8 (Proving Logical Equivalence of Two Propositions)

Determine if $(p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$

$(p \implies r) \wedge (q \implies r)$ seems more complicated, so we'll start with that

$$\begin{aligned}(p \implies r) \wedge (q \implies r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{(Convert Conditionals)} \\ &\equiv (\neg p \wedge \neg q) \vee r && \text{(Un-Distribute)} \\ &\equiv \neg(p \vee q) \vee r && \text{(DeMorgan's Law)} \\ &\equiv (p \vee q) \implies r && \text{(Convert back to Conditional)}\end{aligned}$$

$$\therefore (p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$$

1.4 Section 1.4 - Propositional Functions and Quantifiers

First, let's start with an example

Example 9

Is $x + 3 = 10$ a proposition?

Solution:

No, since x is not known

However, if x is known, then $x + 3 = 10$ is a proposition

The truth value depends on the value of x

Definition 6 (Propositional Function)

We call $x + 3 = 10$ a **propositional function** since the truth value of the proposition depends on the value of x

Example 10 (Examples of Propositional Functions)

For the given propositional functions, determine the truth value for the given value of variable(s)

1. $P(x) : x + 3 = 10$

(a) $P(6)$

Solution:

FALSE since $6 + 3 \neq 10$

(b) $P(7)$

Solution:

TRUE since $7 + 3 = 10$

2. $P(a, b, c) : a^2 + b^2 = c^2$

(a) $P(11, 60, 61)$

Solution:

TRUE

$$11^2 + 60^2 = 121 + 3600 = 3721$$

$$61^2 = (60 + 1)^2 = 3600 + 120 + 1 = 3721$$

(b) $P(147, 201, 513)$

Solution:

FALSE

$$147^2 + 201^2 = \text{odd} + \text{odd} = \text{even}$$

$$513^2 = \text{odd} \neq \text{even}$$

Example 11 (Examples of Notation) 1. Is "there exists a real number x such that $x + 3 = 10$?" a proposition?

Solution:

Yes, and this proposition is **TRUE** since $x = 7$ works

We can also rewrite this as:

$$\exists x \in \mathbb{R} \mid x + 3 = 10$$

2. Is "for all real numbers x , $x + 3 = 10$ " a proposition?

Solution:

Yes, but the proposition is **FALSE** since $x = 6$ doesn't work

We can also rewrite this as

$$\forall x \in \mathbb{R} \mid x + 3 = 10$$

Definition 7 (Quantifiers)

The **existential quantifier** $\exists$ means "there exists"

The **universal quantifier** $\forall$ means "for all" or "for every"

Example 12 (Examples of Quantifiers)

Write each proposition using quantifiers, and determine their truth values

1. There exists an integer x such that $x^2 + 2 > 3x$

Solution:

$$\exists x \in \mathbb{Z} \mid x^2 + 2 > 3x$$

TRUE since $x = 3$ gives $9 + 2 > 9$

2. For all real numbers x , $x^2 + 2 > 3x$

Solution:

$$\forall x \in \mathbb{R} \mid x^2 + 2 > 3x$$

FALSE since $x = 1$ gives $1 + 2 > 3$

Note

We only need one counter example to disprove a proposition that has $\forall$

3. There exists an integer x such that $x^3 = 31687$

Solution:

$$\exists x \in \mathbb{Z} \mid x^3 = 31687$$

FALSE since $\sqrt[3]{31687} \notin \mathbb{Z}$

Note

We needed to verify that every integer does not satisfy the proposition in order to prove it

The negations of each quantifier are as follows:

$$\neg \forall \equiv \exists$$

$$\neg \exists \equiv \forall$$

Example 13 (Negation of Quantifiers)

Give the negation of the following propositions:

1. $\forall x \in \mathbb{R} \mid x^2 + 1 > 0$ **TRUE**

Solution:

$$\exists x \in \mathbb{R} \mid x^2 + 1 \leq 0$$
 FALSE

2. $\exists a \in \mathbb{Z} \mid a^3 + 2 = 3a$ **TRUE** ($a = 1$)

Solution:

$$\forall a \in \mathbb{Z} \mid a^3 + 2 \neq 3a$$
 FALSE

3. For every real number x , $x^2 + 30 < 11x$ **FALSE** ($x = 1$)

Solution:

$$\exists x \in \mathbb{R} \mid x^2 + 30 \geq 11x$$
 TRUE

4. For every prime integer $p > 3$, $p^2 - 1$ is divisible by 24 **TRUE**

Solution:

There exists a prime integer $p > 3$, $p^2 - 1$ is not divisible by 24 **FALSE**

Work

$$p^2 - 1$$

We want $p^2 - 1$ divisible by both 8 and 3

$$(p^2 - 1) = (p - 1)(p + 1)$$

$$(p - 1)(p + 1) = (\text{even})(\text{even})$$

- p is prime
- $p > 3$
- p is odd

Example 14

Statement	When TRUE	When FALSE
$\forall x, P(x)$	$P(x)$ is TRUE for all x	There exists at least one x such that $P(x)$ is FALSE
$\exists x, P(x)$	There exists at least one x such that $P(x)$ is TRUE	For all x such that $P(x)$ is FALSE

Example 15 (Examples with Negation)

Statement	Equivalent	When TRUE	When FALSE
$\neg \forall x, P(x)$	$\exists x, \neg P(x)$	There is at least one x such that $P(x)$ is FALSE	For all x , $P(x)$ is TRUE
$\neg \exists x, P(x)$	$\forall x, \neg P(x)$	$P(x)$ is FALSE for all x	There exists at least one x such that $P(x)$ is TRUE

Example 16 (Examples of Rewriting with Quantifiers)

Rewrite the following with Quantifiers

1. For all positive integers, either $6n + 1$ or $6n - 1$ is prime

Solution:

$\forall n \in \mathbb{Z}^+, 6n - 1$ is prime $\vee 6n + 1$ is prime

FALSE since if $n = 20$ then $6n - 1 = 119$ and $6n + 1 = 121$ are **composite** (not prime)

2. There exists an integer n such that $n^2 + n + 41$ is composite

Solution:

$\exists n \in \mathbb{Z}, n^2 + n + 41$ is composite

TRUE since if $n = 41$ then

$$\begin{aligned} n^2 + n + 41 &= 41^2 + 41 + 41 \\ &= 41(41 + 1 + 1) \end{aligned}$$

which is composite

1.5 Section 1.5 - Nested Quantifiers

Let $\mathbb{P}$ be the set of prime integers

$$\mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$$

With this in mind, we can write the following in two ways:

Consider the statement "for all positive integers n , $6n - 1$ or $6n + 1$ is prime"

1. $\forall n \in \mathbb{Z}^+, 6n - 1 \in \mathbb{P} \vee 6n + 1 \in \mathbb{P}$
2. $\forall n \in \mathbb{Z}^+ \exists p \in \mathbb{P}, 6n - 1 = p \vee 6n + 1 = p$

Example 17 (Example of Nested Quantifiers) 1. For all real x there exists a real number y such that $xy = 1$

$$\forall x \in \mathbb{R} \exists y \in \mathbb{R}, xy = 1$$

FALSE, since if $x = 0$, then $xy = 0 \neq 1$

2. For all positive real x , there exists a real number y such that $xy = 1$

$$\forall x \in \mathbb{R}^+ \exists y \in \mathbb{R}, xy = 1$$

TRUE since if $x \in \mathbb{R}^+$, then $x > 0$, and if $y = \frac{1}{x}$, then $xy = x \frac{1}{x} = 1$

Note

Given x , we can find y

y depends on x

But what happens if we switch $\forall$ and $\exists$?

3. $\exists y \in \mathbb{R} \forall x \in \mathbb{R}^+, xy = 1$

FALSE since if $x = 1$, then $y = 1$, but if $x = 2$, then $y = \frac{1}{2}$, and $1 \neq \frac{1}{2}$

Note

There is a single y that works **FOR ALL** x

y does not depend on x

4. For all positive integers a and b , we have $ab > 0$

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+, ab > 0$$

Note

This is the same as $\forall a, b \in \mathbb{Z}^+$

TRUE since $a, b \in \mathbb{Z}^+$, then $a > 0$ and $b > 0$, hence $ab > 0$

5. There exists integers a , and b such that $a^2 + b^2 = 464$

$$\exists a \in \mathbb{Z} \exists b \in \mathbb{Z}, a^2 + b^2 = 464$$

$$\exists b \in \mathbb{Z} \exists a \in \mathbb{Z}, a^2 + b^2 = 464$$

TRUE $a = 8, b = 20,$

$$\begin{aligned} a^2 + b^2 &= 8^2 + 20^2 \\ &= 64 + 400 \\ &= 464 \end{aligned}$$

Note

The order of the **SAME** quantifier doesn't matter, only for **DIFFERENT** quantifiers does it matter

Example 18

Rewrite the following using Quantifiers

1. "There exists positive integers a and b such that $a^2 + b^2 = 15625$ "

(a) Write as Quantifier

$$\exists a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+, a^2 + b^2 = 15625$$

(b) Write the negation

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+, a^2 + b^2 \neq 15625$$

Work

$$a^2 + b^2 = 5^6$$

$$a^2 + b^2 = 5^4 \cdot 5^2$$

$$3^2 + 4^2 = 5^2$$

$$5^4(3^2 + 4^2) = 5^4 \cdot 5^2$$

$$5^4 \cdot 3^2 + 5^4 \cdot 4^2 = 5^6$$

$$(5^2 \cdot 3)^2 + (5^2 \cdot 4)^2 = 5^6$$

$$(75)^2 + (100)^2 = 5^6$$

$$\therefore a = 75, b = 100$$

TRUE

There's more examples but I'm just too lazy to write them if you get the chance, they're on page 3 of your notes

1.7 Section 1.7 Intro to Proofs

What do we prove? Usually that some pattern or formula is true.

Example 19

$$\begin{aligned}
6 + 8 &= 14 \\
12 + 16 &= 28 \\
2 + 2 &= 4
\end{aligned}$$

We believe that the sum of 2 even integers is even.

Try to prove that: even + even = even

Work An even number can be defined as $2n$, where n is an integer

$$2n + 2m = 2(n + m)$$

Proof. let a, b be even integers.

Then there exists integers n, m such that $a = 2n, b = 2m$

$$a + b = 2n + 2m = 2(m + n)$$

is even, as desired

□

But writing it this way is too long. Let's define some notation:

1.7.1 Set Notation

Name	Symbol	Set Notation
Set of all natural numbers	$\mathbb{N}$	$\{0, 1, 2, 3, \dots\}$
Set of all integers	$\mathbb{Z}$	$\{0, -1, 1, -2, 2, \dots\}$
Set of positive integers	$\mathbb{Z}^+$	$\{1, 2, 3, \dots\}$
Set of negative integers	$\mathbb{Z}^-$	$\{-1, -2, -3, \dots\}$
Set of rational numbers	$\mathbb{Q}$	$\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$
Set of real numbers	$\mathbb{R}$	Limit of all convergent sequences of real numbers
Set of all complex numbers	$\mathbb{C}$	$\{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$

$\exists$ means there exists

Let's rewrite the previous example using this new notation

$a, b \in \mathbb{Z}$, both even

Then, $\exists m, n \in \mathbb{Z}$ such that

$$a = 2m, b = 2n$$

Their sum $a + b = 2m + 2n = 2(m + n)$
is even, as desired

1.7.2 Methods of Proving $p \implies q$

1. **Direct Proof:**

Start with p and *math* then conclude q

2. **Indirect Proof:**

Prove the contrapositive proof $\neg q \implies \neg p$

3. **Finite Cases:**

Use the fact that $p \equiv p_1 \vee p_2 \vee \dots \vee p_n$

Then prove $(p_1 \implies q) \wedge (p_2 \implies q) \wedge \dots \wedge (p_n \implies q)$